

first

$$\theta = (\underset{\uparrow \theta_0}{0.4}, \underset{\uparrow \theta_1}{0.267}, \underset{\uparrow \theta_2}{0.3})$$

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$$

$$h(\theta) = \underline{0.4} + \underline{0.267} x_1 + \underline{0.3} x_2$$

	x_0	x_1	x_2	$h_{\theta}(x) \leftrightarrow y$	error
→	1	1	0	0.667 3	5.44
→	1	0	1	0.7 4	10.89
→	1	1	1	0.967 5	16.27

$$J(\theta) = \frac{1}{2 \times 3} (5.44 + 10.89 + 16.27)$$

$$= \underline{5.43}$$

After 1st iteration...

$$J_1(\theta) = 5.43$$

2nd iteration:

$$\theta = (\underline{0.4}, \underline{0.267}, \underline{0.3})$$

$\downarrow \theta_0 \quad \downarrow \theta_1 \quad \downarrow \theta_2$

$$h(\theta) = \underline{0.4} + \underline{0.267} x_1 + \underline{0.3} x_2$$

x_1	x_2	y	$h(\theta)$	$(y - h(\theta))$
1	0	3	0.667	2.33
0	1	4	0.7	3.3
1	1	5	0.967	4.033

$$\theta := \theta + \alpha \sum_{i=1}^m (y^{(i)} - h_{\theta}(x^{(i)})) x^{(i)}$$

$$\theta_0 = \underline{0.4} + \underline{0} \times \frac{1}{3} (1 \times 2.33 + 1 \times 3.3 + 1 \times 4.033)$$

$$1.00 = 0.00 + 1 \times 4.033$$

$$= 0.72$$

$$\theta_1 = 0.267 + 0.1 \times \frac{1}{3} [1 \times 2.33 + 0 \times 3.3 + 1 \times 4.033]$$

$$= 0.479 \approx 0.48$$

$$\theta_2 = 0.3 + 0.1 \times \frac{1}{3} [1 \times 3.3 + 1 \times 4.033]$$

$$= 0.544$$

$$h(\theta) = 0.72 + 0.48x_1 + 0.54x_2$$

x_1	x_2	y	$h_\theta(x)$	(error)
1	0	3	1.2	3.24
0	1	4	1.27	7.48
1	1	5	1.75	10.6

$$\rightarrow (y - h(\theta))^2$$

$$J(\theta) = \frac{1}{2m} \sum (\text{error})^2$$

$$= \frac{1}{2 \times 3} \times (3.24 + 7.48 + 10.6)$$

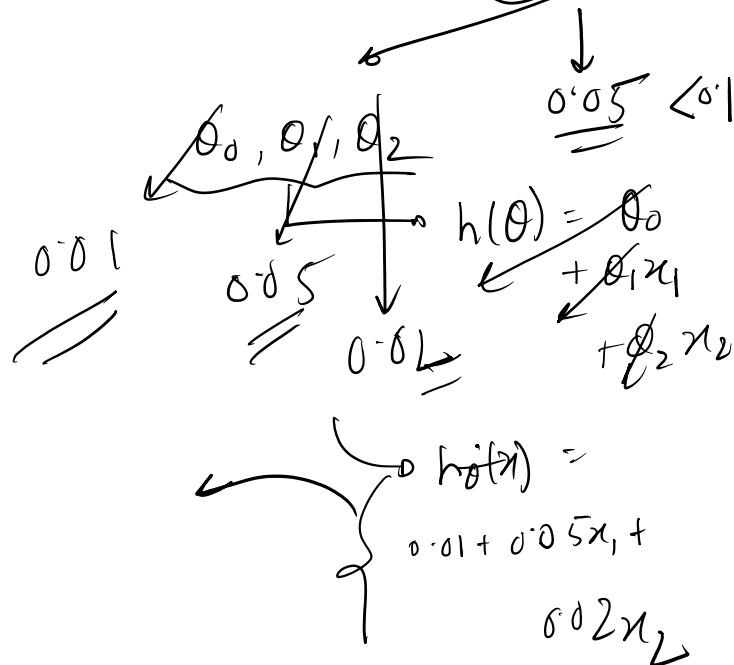
$$= 3.55$$

$$J_2(\theta) = 3.55$$

$$5.43 \rightarrow 3.55$$

$$(J_n(\theta)) - (J_{n-1}(\theta)) < 0.1$$

$i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_{100} \rightarrow i_{101}$



given $\rightarrow$ (m) training samples

Find the optimum hypothesis so that it fits mostly with the given training data. Use batch gradient descent to update the parameters of hypothesis.

Simulation